

The Link Function Problem in Psychological Interaction Testing

Margherita Calderan¹, Filippo Gambarota², Laura Sità¹, Tommaso Feraco¹, and Enrico Toffalini¹

¹Department of General Psychology, University of Padova

²Department of Developmental Psychology and Socialization, University of Padova

Author Note

Margherita Calderan  <https://orcid.org/0009-0005-5668-5162>

Filippo Gambarota  <https://orcid.org/0000-0002-6666-1747>

Laura Sità  <https://orcid.org/0009-0000-6017-9979>

Tommaso Feraco  <https://orcid.org/0000-0002-8920-5330>

Enrico Toffalini  <https://orcid.org/0000-0002-1404-5133>

Correspondence concerning this article should be addressed to Enrico Toffalini,
Department of General Psychology, University of Padova, Via Venezia 8, Padova 35131, Italy,
Email: enrico.toffalini@unipd.it

Abstract

Psychologists routinely test interactions, and whether an interaction exists depends on the scale on which effects are assumed to add. In psychology, the observed score and the construct one actually wants to measure are generally on different scales, and equal differences on one scale are unequal differences on another. Because an interaction is a difference between differences, the same data can show an interaction on one scale and none on another. We frame this scale problem in the language of generalized linear models, where the outcome family describes the response and the link function fixes the scale on which the presence of an interaction is defined. The paper has two connected parts. First, a preregistered review of recent articles from five leading psychology journals shows that interaction testing is pervasive, that the link function is usually left unspecified, and that constrained outcomes are routinely analyzed on the wrong scale. Second, worked examples and simulations develop three common response formats, namely forced-choice accuracy with a chance floor, within-family link choices such as logit versus probit, and sum scores as a measurement case, and quantify how a plausible-looking link can turn an additive structure on one scale into a statistically significant interaction on another. A final diagnostic simulation shows that residual checks, a link test, and AIC comparisons give uneven protection and do not by themselves decide which scale the scientific claim concerns. Interaction claims remain underspecified until researchers state, justify, and probe the scale on which no interaction is assumed.

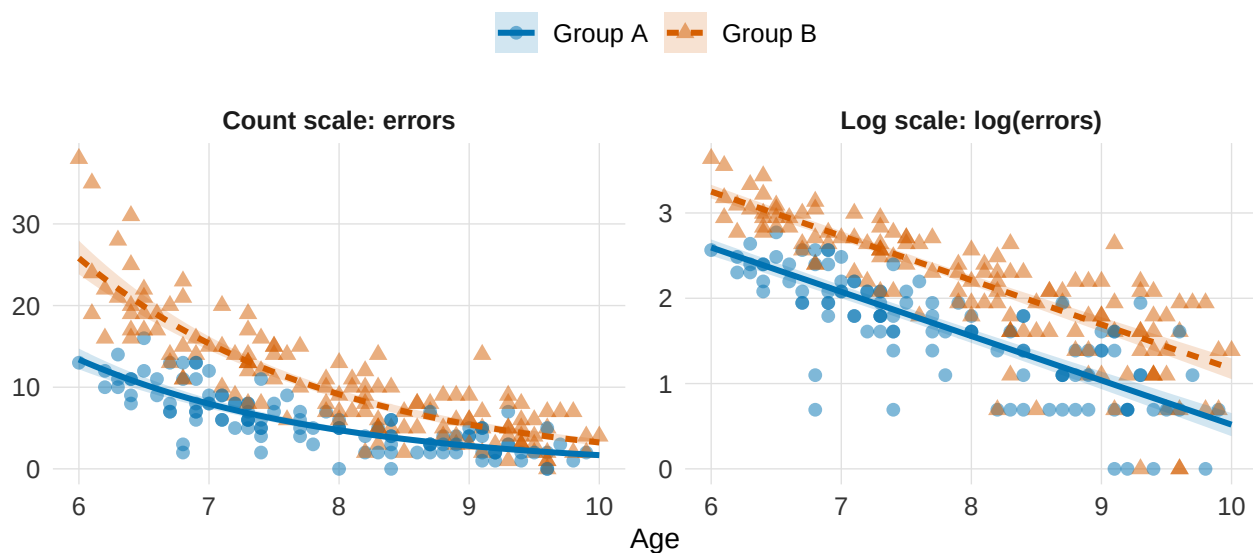
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The Link Function Problem in Psychological Interaction Testing

Suppose two groups of children differ in their overall error rates on a task, and errors decrease with age. Figure 1 shows simulated data from such a scenario. In the left panel, errors decline with age in both groups, and the difference between groups is large at age six and nearly closed at age ten. Is there an age-by-group interaction here?

Figure 1

Simulated error counts for two groups of children aged 6 to 10, shown on the observed count scale (left) and on the log scale (right). The two groups have exactly the same age effect on the log of the expected error count, so on the log scale the two fitted lines are parallel, with a constant group gap at every age. Mapped back onto the count scale, the same model produces converging curves, so the group difference shrinks with age. Whether this pattern is an interaction depends on the scale on which effects are assumed to add: on the count scale the group difference changes with age; on the log scale it does not.



Because an interaction is a difference between differences, the question is whether the group difference itself changes with age. On the observed count scale, the group effect does appear to shrink as age increases, so it looks like an interaction. But looking at the log-scale (Figure 1, right panel), both groups have the same age slope, meaning their fitted lines are parallel. A constant difference on the log scale corresponds to a constant ratio on the count scale, so the

absolute gap between groups is proportional to the baseline and shrinks as errors decline with age. So whether we call this an interaction is not a fact about the data. It depends on which scale we assume effects add up linearly on. Additive on the log scale means multiplicative (and non-parallel) on the count scale, and vice versa.

This ambiguity extends far beyond this single illustration, as interaction claims are central to psychological theory [ADD REF]. Whenever researchers ask whether an effect is moderated, or whether it holds only under certain boundary conditions, they are asserting that the impact of one variable depends on another. This count example therefore demonstrates that the reality of such a claim can shift entirely based on the scale on which those differences are computed, as equal differences on one scale are different differences on another scale.

In psychology, this scale dependence has a characteristic source in measurement. Researchers routinely conceptualize constructs such as cognitive ability or symptom severity as continuous latent dimensions [ADD REF]. Those dimensions are rarely observed directly; what reaches the data are bounded, discrete quantities, such as counts, ratings, and accuracy scores that carry floors, ceilings, and chance levels. Each measurement scale imposes its own mapping from the latent dimension to the observed response, and that mapping fixes the scale on which effects add.

Generalized linear models (GLMs) make this mapping explicit and give it a standard vocabulary (McCullagh & Nelder, 1989; Nelder & Wedderburn, 1972). A GLM separates two choices that are easy to conflate: the outcome family, which describes the response, and the link function, which sets the scale on which effects combine (Box 1).

Box 1: Generalized linear models: outcome family and link function

The *outcome family* is the distribution assumed for the response. It defines which values the outcome can take and how observations vary around their expectation. The *link function*, $g(\cdot)$, maps the expected outcome, $\mu_i = E(Y_i)$, onto the linear predictor, η_i , the scale on which the

model is additive. With two predictors and their interaction term,

$$g(\mu_i) = \eta_i, \quad \eta_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 X_i Z_i.$$

The inverse link, $g^{-1}(\cdot)$, maps this additive structure back onto the expected response. Its familiar role is to keep expected values within the admissible range of a constrained outcome: $\exp(\eta)$ keeps expected counts positive, and the inverse logit keeps probabilities between 0 and 1. Less obviously, the link thereby defines the scale on which effects add, and therefore what the absence of an interaction, $\beta_3 = 0$, means in the fitted model. The product-term coefficient β_3 is therefore an interaction on the link scale. A response-scale interaction is a different quantity: a difference between expected differences on the observed outcome, for example, whether the group difference in expected errors changes with age in Figure 1. Unless the link is the identity, the two quantities can disagree (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019). Each family has a canonical link, but other links are legitimate. A Gaussian linear model is the special case that combines a Gaussian family with an identity link, so the additive scale and the response scale coincide and the scale question is invisible. The model behind Figure 1 is a Poisson family with a log link. The family respects that errors are non-negative counts, and the link places additivity on the log scale.

Often the outcome family is chosen with care while the link function is left at its software default, fixing the scale of additivity without deliberation. Two models can share a family and still place additivity on different scales. For example, a Gamma family suits reaction times data; within that family, a log link makes effects additive in log milliseconds, a proportional change, and the canonical inverse link makes them additive in reciprocal means, a change in speed. A binomial family fits correct-versus-incorrect responses and leaves the link open among logit, probit, and chance-corrected forms.

While the family restricts the set of admissible links, model estimates depend heavily on both choices (Hainmueller et al., 2019; Rimpler et al., 2026). When the outcome distribution or

the functional form is misspecified, additive structure can appear as a spurious interaction (Ai & Norton, 2003; Busemeyer & Jones, 1983; Czado & Santner, 1992; Domingue et al., 2024; Liddell & Kruschke, 2018; Lubinski & Humphreys, 1990; McCabe et al., 2022; Mize, 2019; Yu & Huang, 2019), and measurement error and low power compound the problem (Baranger et al., 2023; Castillo et al., 2026). That said, an interaction on a scale other than the one implied by the generative model is not a false positive in a strict statistical sense; it is a genuine mathematical property of that scale (Cox, 1984; Domingue et al., 2024; Loftus, 1978; McCabe et al., 2022). Some interactions are invariant to scale; crossover interactions, for instance, survive any monotonic rescaling of the response variable (Cox, 1984; Loftus, 1978; Wagenmakers et al., 2012).

This article addresses these issues in two parts. The first is a preregistered descriptive review of recent empirical articles in five prominent psychology journals, quantifying how often interactions are tested, how often the outcome family and link function are made explicit, and which response formats predominate. The second develops the link function problem in three common response formats: forced-choice accuracy with a chance floor, sum scores, and within-family link choices such as logit versus probit. Using simulations, it quantifies how often an inadequate link turns a purely additive structure into a statistically significant interaction, and whether standard diagnostics detect the misspecification. Table 4 then summarizes these insights into a practical workflow for applied research.

Current Practice in Psychological Interaction Testing: A Preregistered Review

The review asks whether the scale-dependence problem is visible in current reporting practice: how often psychological articles test interactions, and how often they make the outcome family and link function explicit enough for a reader to know which interaction claim was tested. To our knowledge, it is the first review to quantify link-function reporting in psychological interaction tests. The coding is deliberately descriptive, based only on what a published report makes observable, so it documents reporting practice rather than the correctness of any substantive conclusion.

Review Method

The protocol was preregistered before any record was screened (<https://osf.io/bdu73/>), together with the coding data dictionary. The essentials are given here; the full protocol, sampling frame, and reliability are in the Supplement.

Following the preregistration, we sampled one journal from each of five areas: *Psychological Science* (general), *Journal of Experimental Psychology: General* (experimental), *Journal of Personality and Social Psychology* (social), *Developmental Science* (developmental), and *Psychological Medicine* (clinical). The five were selected on preregistered discretionary criteria: a first-quartile 2023 Impact Factor, a broad rather than narrowly specialized scope, not primarily publishing reviews, and strong reputation in their field, as jointly judged by the coauthors. From each journal's full set of 2024 articles in Scopus, records were screened in a fixed random order until 40 eligible interaction-testing articles were reached, for a preregistered total of 200. Eligible articles were empirical reports of original research; reviews, meta-analyses, editorials, theoretical and qualitative-only articles, replication-only studies, and psychometric validation studies were excluded. Detailed coding was applied only to articles testing at least one interaction, and only to observed-outcome analyses: articles whose relevant tests were purely latent-variable or structural-equation models fell outside the coding scheme (12 such exclusions).

Each interaction-testing article was independently coded by two researchers, with disagreements resolved by discussion or by consulting the other coauthors. Generative AI tools were used only as an optional aid to locate passages within an article; all coding used in the review is human.

For each article, the coders recorded five things: whether at least one tested interaction used a non-identity link; whether the link function was explicitly reported; whether at least one interaction applied an identity-link analysis to a constrained observed outcome, which we call a *formally incorrect identity-link analysis*; whether at least one such interaction was statistically significant, or interpreted analogously in a Bayesian analysis; and which response-variable types were analyzed. By a formally incorrect identity-link analysis we mean a Gaussian linear

regression with an identity link applied to an outcome with formal constraints, such as bounds, discreteness, positivity, or a known chance level. This is a conservative screening flag, applied only when the constraint was identifiable from the report; it is not an error rate and does not imply the substantive conclusion is wrong. The full operational rules the coders followed, including borderline cases, are tabulated in the Supplement. Interrater agreement on the double-coded records was high (Cohen's κ between 0.66 and 0.81); per-variable agreement and coder margins are reported in the Supplement.

Review Findings

First, interaction testing was common. The review identified 331 eligible empirical articles, and 200 of these articles tested at least one interaction, corresponding to 60.4%, 95% CI [55.1, 65.5]. This confirms that interaction testing is a routine part of empirical reporting in the sampled journals.

Second, model-scale reporting was uncommon. Among the 200 interaction-testing articles, 58 used at least one non-identity link function, 29%, 95% CI [23.2, 35.6], and only 46 explicitly reported the link function, 23%, 95% CI [17.7, 29.3]. Thus, in about 77.0% of interaction-testing articles, the link function was not explicitly stated.

Even the 46 explicit cases rest on a generous coding rule. We counted a link-specific model name as an explicit declaration, because such names identify the link: “logistic regression” names the logit link, “probit regression” the probit, and “Poisson regression” the log. Naming only a family or a “generalized linear model” was not enough. Most of the explicit cases arose this way, from the model's name rather than from a statement of the link in words; had we required the link to be named directly, the count would be lower still.

Third, constrained outcomes were often analyzed on the identity scale. Among the 200 interaction-testing articles, 144 included at least one formally incorrect identity-link analysis of a constrained observed outcome, 72%, 95% CI [65.4, 77.8]. As noted, this is a conservative screening count, flagged only when the constraint was clear from the report. Of these 144 cases, 124 reported at least one statistically significant interaction, 86.1%, 95% CI [79.5, 90.8]; that is,

62.0% of all interaction-testing articles combined a constrained outcome, an identity-link analysis, and a significant interaction claim. This is the same concern raised by work on outcome-model mismatch in interaction testing (Domingue et al., 2024; Liddell & Kruschke, 2018). Table 1 summarizes these results.

Table 1

Descriptive summary of the preregistered review. Formally incorrect identity-link analyses of constrained observed outcomes are cases where the reported outcome had formal constraints that make identity-link additivity formally incorrect as a generative model unless explicitly justified as an approximation or as the target observed-score metric.

Quantity	n	Denom.	%	95% CI
Eligible empirical articles	331	331	100.0%	[98.9%, 100.0%]
Eligible empirical articles testing at least one interaction	200	331	60.4%	[55.1%, 65.5%]
Interaction-testing articles using at least one non-identity link	58	200	29.0%	[23.2%, 35.6%]
Interaction-testing articles explicitly reporting the link function	46	200	23.0%	[17.7%, 29.3%]
Interaction-testing articles with formally incorrect identity-link analyses of constrained observed outcomes	144	200	72.0%	[65.4%, 77.8%]
Formally incorrect identity-link cases with at least one significant interaction	124	144	86.1%	[79.5%, 90.8%]
Eligible empirical articles not testing interactions	131	331	39.6%	[34.5%, 44.9%]

The pattern was present across the sampled journals. In the supplementary by-journal table, the proportion of formally incorrect identity-link analyses of constrained observed outcomes among interaction-testing articles ranged from 40.0% in Psychological Medicine to 95.0% in Psychological Science. In this sample, Psychological Medicine had the lowest screening proportion and the highest non-identity-link proportion (40.0%). This journal, which uses many outcomes for which non-identity links are standard, contributed the fewest formally incorrect identity-link cases.

The outcomes used in interaction analyses were heterogeneous. The most common broad,

non-exclusive categories were binary, accuracy, or proportion outcomes (84 articles), sum scores or composites (66 articles), ordinal, Likert, or rating outcomes (36 articles), response times or durations (30 articles), neural or physiological measures (23 articles), difference or distance scores (17 articles), counts or error counts (9 articles), and correlations or associations (8 articles). This heterogeneity matters because the identity link has different implications across outcome types. It also shows why the worked examples focus on bounded, binomial, ordinal, and aggregated responses. Many observed psychological variables are built from binary or ordered response events, then reported as proportions, ratings, averages, totals, or composites. For some approximately continuous outcomes, additivity on the observed scale may be a defensible target. For bounded, discrete, ordinal, chance-constrained, or heavily transformed outcomes, however, additivity on the observed scale is an assumption that should usually be stated and examined. This is especially clear for ordinal and Likert-type outcomes, where treating ordered categories as metric can affect estimates, error rates, and interaction conclusions ([Bürkner & Vuorre, 2019](#); [Liddell & Kruschke, 2018](#)).

Table 2

Broad response-variable categories in interaction-testing articles. Categories are non-exclusive, so one article can contribute to more than one row.

Outcome type	n	Denominator	Percent
Binary / accuracy / proportions	84	200	42.0%
Sum scores / composites	66	200	33.0%
Ordinal / Likert / ratings	36	200	18.0%
Response times / durations	30	200	15.0%
Counts / error counts	9	200	4.5%
Neural / physiological measures	23	200	11.5%
Correlations / associations	8	200	4.0%
Difference / distance scores	17	200	8.5%

What the Review Shows, and What It Does Not

As the first review to quantify link-function reporting in psychological interaction tests, it yields three clear results. First, interaction testing is pervasive: 60% of eligible empirical articles

tested at least one interaction. Second, the scale of additivity is usually left implicit: only 23% named the link function, and even that count leans on counting model names such as “logistic regression” as explicit. Third, constrained outcomes are routinely analyzed on the identity scale: 72% of interaction-testing articles applied an identity-link analysis to an outcome with formal constraints, and most of these reported a significant interaction. In leading journals, then, interaction tests are common while the scale on which no interaction is assumed is rarely stated.

The review documents how analyses were reported. A descriptive review cannot establish how often published interactions are scale artifacts, because that judgment requires knowing the scale on which the data-generating process is additive. The remainder of the paper therefore turns to settings where the generating process is known by construction.

The Link Function Problem in Three Common Response Formats

To demonstrate the practical consequences of link misspecification, we develop three complementary simulation studies:

1. *Forced-Choice Accuracy*: This case holds the binomial outcome family fixed and evaluates whether the link function must explicitly encode a task-implied chance floor (e.g., a 0.50 guessing threshold).
2. *Within-Family Link Selection (Logit vs. Probit)*: This case holds both the binary family and the 0-to-1 bounds fixed, testing whether choosing between these two standard link functions alters the results.
3. *Sum Scores*: This shows how adding up individual item responses to create a bounded total score blends two distinct issues: how individual items map onto a total scale, and how that scale alters what it means for an effect to be truly additive.

Each case develops the conceptual problem and then reports a simulation that quantifies it. Across the three cases, the simulations address two questions: first, whether a purely additive data-generating process on one scale is detected as a statistically significant interaction when modeled on a different scale; and second, whether standard diagnostics reliably identify the

misspecified link. The simulation scenarios were chosen to represent plausible psychological data patterns. Each reproduces a common interaction-testing situation: two effects that are already well established, for example a group difference combined with a developmental trend or a condition effect, and the question of whether one effect moderates the other. The two main effects were always set strong enough to push expected values toward a floor, ceiling, or chance level while keeping them clearly inside the observed range. This is the region where link curvature acts most strongly, and it corresponds to patterns that psychological studies routinely report, such as accuracy that approaches ceiling in one group and stays near chance in another.

Each simulation applies the same design. Data are generated with no interaction on the generating scale; models assuming additivity on the generating scale and on plausible alternative scales are fitted to the same data; and the proportion of statistically significant interactions across replications quantifies the inferential risk attached to each scale choice. The reported rates therefore quantify this mechanism under known generating processes, and they are specific to these tuned scenarios.

Forced-Choice Accuracy and the Non-Zero Chance Floor

If a participant answers Y_i of k_i trials correctly, a binomial sampling model, $Y_i \sim \text{Binomial}(k_i, p_i)$, is the natural starting point. The family fixes the sampling process; it does not fix the lower bound that is meaningful for the task. In an m -alternative forced-choice task, guessing yields an expected accuracy of $c = 1/m$, which is .50 in a two-alternative or true/false task, so a claim about performance above guessing places the lower asymptote of the performance curve at c , not at 0.

Three specifications treat this floor differently. A Gaussian identity model on the raw proportions ignores the trial process, assumes additivity in percentage-correct units, and can predict outside the 0-to-1 range. A standard binomial logit or probit model, $p_i = F(\eta_i)$ with F the logistic or standard-normal CDF, respects the sampling process and the 0-to-1 bounds and still places the lower asymptote at 0. A chance-corrected binomial link keeps the binomial family and maps the linear predictor onto the interval from chance to 1,

$$p_i = c + (1 - c)F(\eta_i),$$

which for $c = .50$ becomes $p_i = .50 + .50 F(\eta_i)$; equivalently, the link acts on the above-chance quantity $(p_i - c)/(1 - c)$. A interaction term in the standard model tests additivity over the full 0-to-1 range, and a interaction term in the chance-corrected model tests additivity on a performance-above-chance scale that already encodes the task floor.

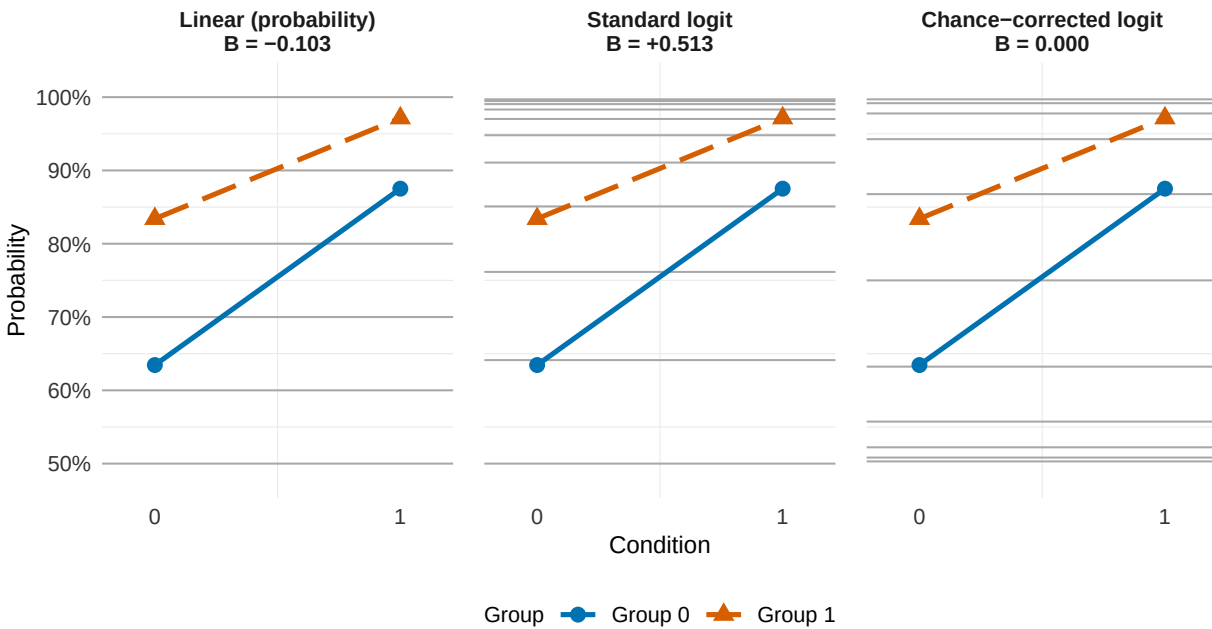
The mismatch is most severe when one group or condition lies near chance. Suppose age has the same effect in two groups on the performance-above-chance scale, with no true age-by-group interaction term there. Near .50 the chance-corrected curve has little response-scale room, so the lower-performing group shows stronger compression on the observed accuracy scale. A standard logit or probit link puts that compression near 0 in place of near .50. The age-by-group difference between differences is exactly zero on the chance-corrected scale and systematically nonzero on the standard logit scale, so the standard model detects a genuine property of its own scale (Figure 2).

The chance-corrected link is most defensible when the chance level is known from the task design, guessing is a plausible lower-bound process, and below-chance performance is not substantively meaningful for the target claim. In the simplest case, the chance level can be fixed by design, for example $c = 0.50$ in a 2-AFC task. In other cases, analysts may estimate the lower asymptote or report sensitivity analyses over plausible values of c . When there are lapses, response biases, item heterogeneity, learning across trials, systematic below-chance performance, or participant-specific guessing strategies, a richer model is often needed instead [ADD REF].

The chance-corrected GLM link used here closely relates to psychometric-function models (Knoblauch & Maloney, 2012; Wichmann & Hill, 2001) that estimate the chance process at the level where it actually operates. For example, the 3-parameter logistic Item Response Theory model has a guessing parameter defined at the item level rather than the design level [REF]. The GLM link should thus be seen as a design-level approximation, not a substitute for item-level measurement models when the data and scientific question call for that additional structure.

Figure 2

A minimal 2×2 design crosses a condition (x-axis) with a group (color), shown on three scales. The four cell probabilities are identical in every panel; only the scale on which the interaction is evaluated changes. The data were generated with no condition-by-group interaction on the chance-corrected logit scale, which has a 0.50 chance floor, so the interaction coefficient is exactly zero on that scale. The same cells imply a positive interaction on the standard logit scale and a negative interaction on the raw probability scale. Light horizontal lines are equally spaced on each panel's own link scale, so equal steps on that scale map to unequal steps on the probability axis.



Simulation 1: Design and Results

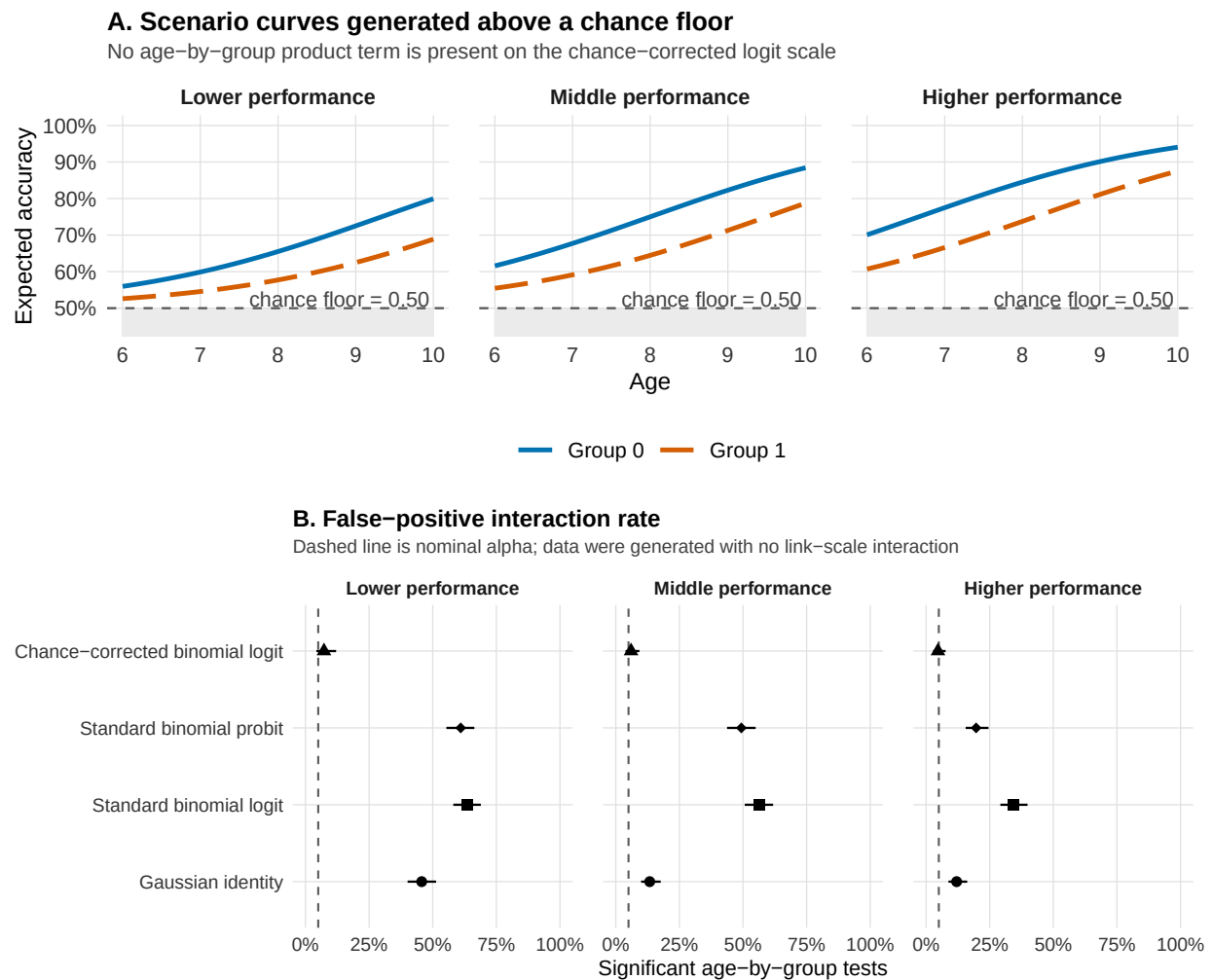
In each of 300 replications we generated accuracy for $N = 250$ participants with $k = 20$ trials each, from a chance-corrected logit model with a .50 floor, a positive age effect, a group difference, and no age-by-group interaction term on the generating scale. Three scenarios differ only in overall performance, set by shifting the intercept: in the *lower-performance* scenario the two groups lie close to the .50 floor (expected accuracies roughly .53 to .80 across ages 6 to 10), in the *middle-performance* scenario they occupy the middle of the above-chance range (roughly .55

to .88), and in the *higher-performance* scenario they approach the ceiling (roughly .61 to .94).

Each simulated dataset was analyzed with four aggregate models, one row per participant: a Gaussian linear model on the observed proportions, standard binomial-logit and binomial-probit GLMs on the correct-response counts, and the chance-corrected binomial-logit GLM that matches the generating scale. We recorded how often each returned a significant age-by-group interaction term at $\alpha = .05$, a false positive by construction. Figure 3 shows the simulation results; the per-scenario rates are tabulated in the Supplement.

Figure 3

Simulation 1: forced-choice accuracy with a task-implied chance floor. Panel A shows the deterministic scenario curves generated by the chance-corrected logit model. The grey region marks values below the .50 chance floor, and the dashed horizontal line marks chance performance. Panel B shows the false-positive rate for the age-by-group interaction term across replications, for each fitted model and performance scenario. Data are generated with no age-by-group interaction on the chance-corrected scale.



The chance-corrected model stays near the nominal .05 in every scenario, with false-positive rates between 5% and 7%, because it matches the generating scale. The misspecified models inflate, most severely where performance lies near the floor: in the lower-performance

scenario the standard logit rejects 64% of the time, the standard probit 61%, and the Gaussian identity 46%, against the nominal 5%. The inflation shrinks as performance moves away from the floor, falling in the higher-performance scenario to 34% for the logit and 20% for the probit, and it persists wherever the inverse link is curved. When the true floor is .50 and the fitted link places it at 0, the interaction term compensates for the wrong no-interaction baseline.

The chance-corrected model can be numerically demanding. Near the floor the likelihood carries little information about changes on the above-chance scale, and some fits fail to converge; in the lower-performance scenario, only 179 of the 300 replications did. This instability is itself informative, signaling that the design gives limited evidence about above-guessing differences in that region. Weakly informative Bayesian priors or item-level models can stabilize estimation, at the cost of added assumptions that should be stated and checked [REF O TOGLIERE FRASE].

Within-Family Link Choice

The forced-choice case involved a link that was plainly wrong for the task, placing the lower asymptote at 0 when the design fixed it at chance. The link-function problem also takes a subtler form, between two links that are both standard for binary data, both respect the 0-to-1 bounds, and usually produce almost indistinguishable fits. Logit and probit give nearly identical fitted probabilities across most of the response range, which is why switching between them leaves most applied conclusions unchanged. That practical equivalence makes the pair a stringent test of the link-function problem, because a distortion of an interaction even here would extend well beyond obvious misspecification. But the equivalence is only local: near the floor, ceiling, chance level, or distributional tails, the two links imply different distances on the link scale, even where their fitted probabilities nearly.

Additionally, the two links still encode different substantive interpretations. A logit link is natural when the interpretation is in odds or odds ratios: additivity on the linear predictor scale then means additivity in log odds, and exponentiated coefficients have a direct odds-ratio interpretation. A probit link is natural when the response can be represented as threshold crossing on an approximately normal latent variable, such as ability, signal strength, response tendency,

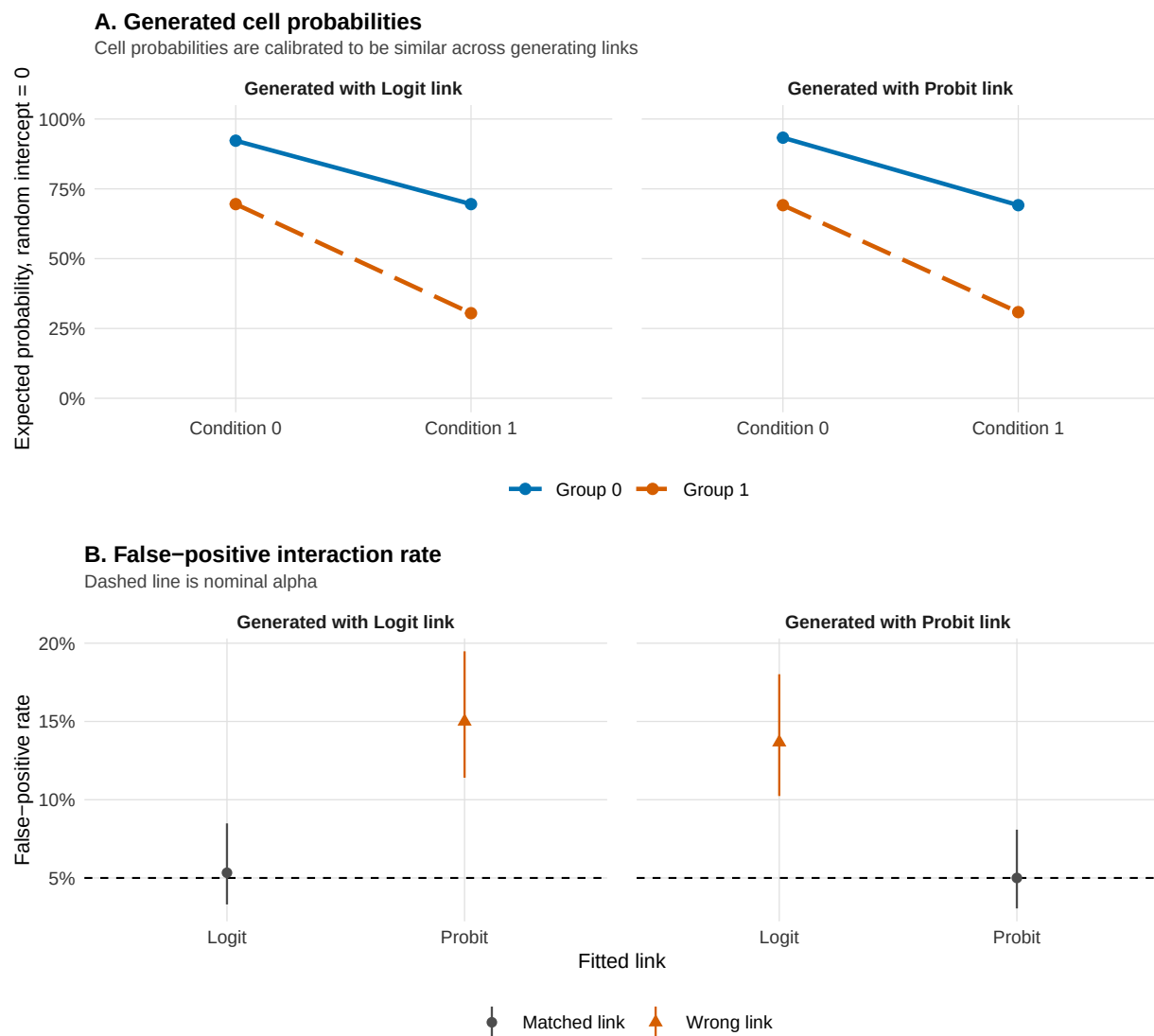
propensity, or latent correctness (Dunn & Smyth, 2018).

Simulation 2: Design and Results

In each of 300 replications we generated repeated binary responses for $N = 700$ participants in a two-group [**un pò surreale come campione? l'abbiamo messo perchè i falsi positivi tra probit e logit si trovano solo quando sia ha una potenza altissima?**], two-condition design, with 15 trials per participant-condition cell and a subject random intercept (latent intraclass correlation 0.30). The condition-by-group interaction term was zero on the generating link scale. We crossed the generating link with the fitted link, using logit and probit in both roles, and fitted every model as a random-intercept GLMM, recording how often each returned a significant condition-by-group interaction term at $\alpha = .05$, a false positive by construction. Figure 4 shows the results; the full crossed-link rates are tabulated in the Supplement.

Figure 4

*Simulation 2: within-family link choice in repeated-trial random-intercept GLMMs. Panel A shows the deterministic cell probabilities generated by additive logit or probit models at random intercept 0. Panel B shows the false-positive rate for the condition-by-group interaction term when the generating and fitted links are crossed. Error bars are Wilson 95% confidence intervals, and the dashed line marks nominal alpha. **COS'è random intercept = 0?***



When the fitted link matches the generating link, the false-positive rate stays near the nominal .05: 5% for logit fitted to logit-generated data and 5% for probit fitted to probit-generated

data. Crossing the links inflates it to 14% when a logit model is fitted to probit-generated data and 15% when a probit model is fitted to logit-generated data, roughly a tripling of the nominal rate. The inflation is smaller than in the forced-choice case, and it appears among links that share the same family, the same 0-to-1 bounds, and nearly overlapping fitted curves in the middle of the probability range, the setting many applied analysts treat as interchangeable. Under the wrong link the same cell probabilities imply a small but deterministic nonzero interaction term on the fitted scale, which repeated sampling detects. A result that is stable across both links is easier to defend; one that appears only under a single link should be reported as link-conditional, or translated into a response-scale estimand and checked across plausible links.

Sum Scores as Bounded and Discrete Outcomes

Sum scores often combine a link-scale issue with a target-metric issue. They may be analyzed as Gaussian identity outcomes, transformed bounded scores, ordinal summaries, or item-level response models, depending on the scientific target and on the information available. We include them because many sum scores aggregate binary or ordinal response processes. For interaction claims, the risk is the same: the chosen metric or link defines the scale on which additivity is tested.

Sum scores make the link-function problem inseparable from measurement. Many psychological outcomes are reported as totals or averages across questionnaire items, task items, symptoms, errors, or ratings. Recent debates on sum scores make the same point from a psychometric angle: a total score can be useful and transparent, but it also encodes assumptions about how item responses represent the target construct ([McNeish, 2023](#); [McNeish & Wolf, 2020](#); [Widaman & Revelle, 2023](#)). These scores are often treated as if they were direct observations of an approximately continuous psychological dimension. The latent construct and the observed score are different objects. The construct may be conceived as a continuous dimension, and may sometimes be reasonably approximated as smooth or roughly normal. The sum score is produced by a finite set of item responses through thresholds, endorsement probabilities, success probabilities, category labels, and scoring rules. It is therefore a remapped version of the target

continuum.

This point matters even when the construct is otherwise well behaved. A normally distributed latent variable can still generate a bounded, discrete, skewed, or non-interval observed score. A total based on binary or ordinal items has only a limited number of possible values. It has a lower and upper bound. It can be skewed when items are too easy, too hard, rarely endorsed, or almost always endorsed. It can be compressed near floor or ceiling, but compression is not restricted to the endpoints. Adjacent raw-score values may correspond to different distances on the latent scale at different regions of the score range. These features are common in psychological variables ([Micceri, 1989](#)), and they are central to why treating ordinal responses as metric can produce false alarms, missed effects, reversals, and misleading interactions ([Liddell & Kruschke, 2018](#)).

Sum scores can be useful, transparent, reliable enough for many purposes, comparable across studies, and pragmatically tied to how instruments are scored in applied settings ([Widaman & Revelle, 2023](#)). A Gaussian identity model for a sum score still makes a scale claim. It assumes that effects are additive in raw-score units on the observed score scale. In an interaction model, the no-interaction baseline is that the expected raw-score effect of one predictor is constant across levels of the other predictor. Sometimes raw-score units are the target estimand. For example, a clinician, teacher, or researcher may care about a two-point difference on the reported score because decisions and interpretations are made on that metric.

Gaussian identity models are more defensible when the score has many possible values, the distribution is not strongly skewed, observations are not concentrated near floor or ceiling, residual patterns are not strongly tied to fitted values, and the substantive claim is explicitly about raw-score differences. They are less defensible when the score has few levels, is built from ordinal item responses with unclear spacing, shows floor or ceiling concentration, or is interpreted as evidence about moderation on a latent construct scale.

The problem becomes sharper when the verbal claim moves from the manifest score to the underlying construct. A group near the upper bound may show a smaller observed treatment effect

simply because the score has less room to increase, with the treatment equally effective on the construct. A group located near a dense region of thresholds may show a larger raw-score change because the instrument is more sensitive there. The interaction term can then partly reflect the target metric rather than genuine moderation in the psychological process. A closely related problem has been shown in item-level work on genotype-by-environment interactions, where summed item scores can mask genuine interactions or generate spurious ones through scale properties of the observed score ([Molenaar & Dolan, 2014](#)). This is the sum-score version of outcome-model misspecification and target-metric mismatch ([Domingue et al., 2024](#)).

Several pragmatic alternatives can make the scale assumption more explicit. For a manifest sum score, a Gaussian identity model can be a defensible approximation when the claim is explicitly about raw-score units, the score has many levels, predicted values remain in range, and floor or ceiling compression is limited. When the claim is closer to an underlying construct, the better solution is an explicit measurement model, such as item-level ordinal, IRT-inspired, or SEM models. A bounded-score probit model can be used only as a sensitivity analysis on the manifest total. It is not a substitute for a measurement model. This point is also consistent with the sum-score debate: replacing a total score with a latent-score or item-level model moves the assumptions into the measurement model ([McNeish, 2023](#); [McNeish & Wolf, 2020](#); [Widaman & Revelle, 2023](#)). The role of the alternatives considered here is not to establish the true scale, but to test whether the interaction conclusion depends on the manifest sum-score metric ([Bürkner & Vuorre, 2019](#); [Domingue et al., 2024](#); [Liddell & Kruschke, 2018](#)).

The recommendation is to make the assumed scale visible. Researchers can inspect the score distribution, plot predicted values near the bounds, report response-scale contrasts, and compare Gaussian identity results with bounded-score probit, ordinal, bounded-count, or item-level alternatives when these are plausible. Stable interactions across these specifications are easier to defend. Interactions that appear only on the manifest sum-score metric can still be informative, but they should be framed as properties of that score and interpreted as construct-level moderation only with supporting measurement assumptions.

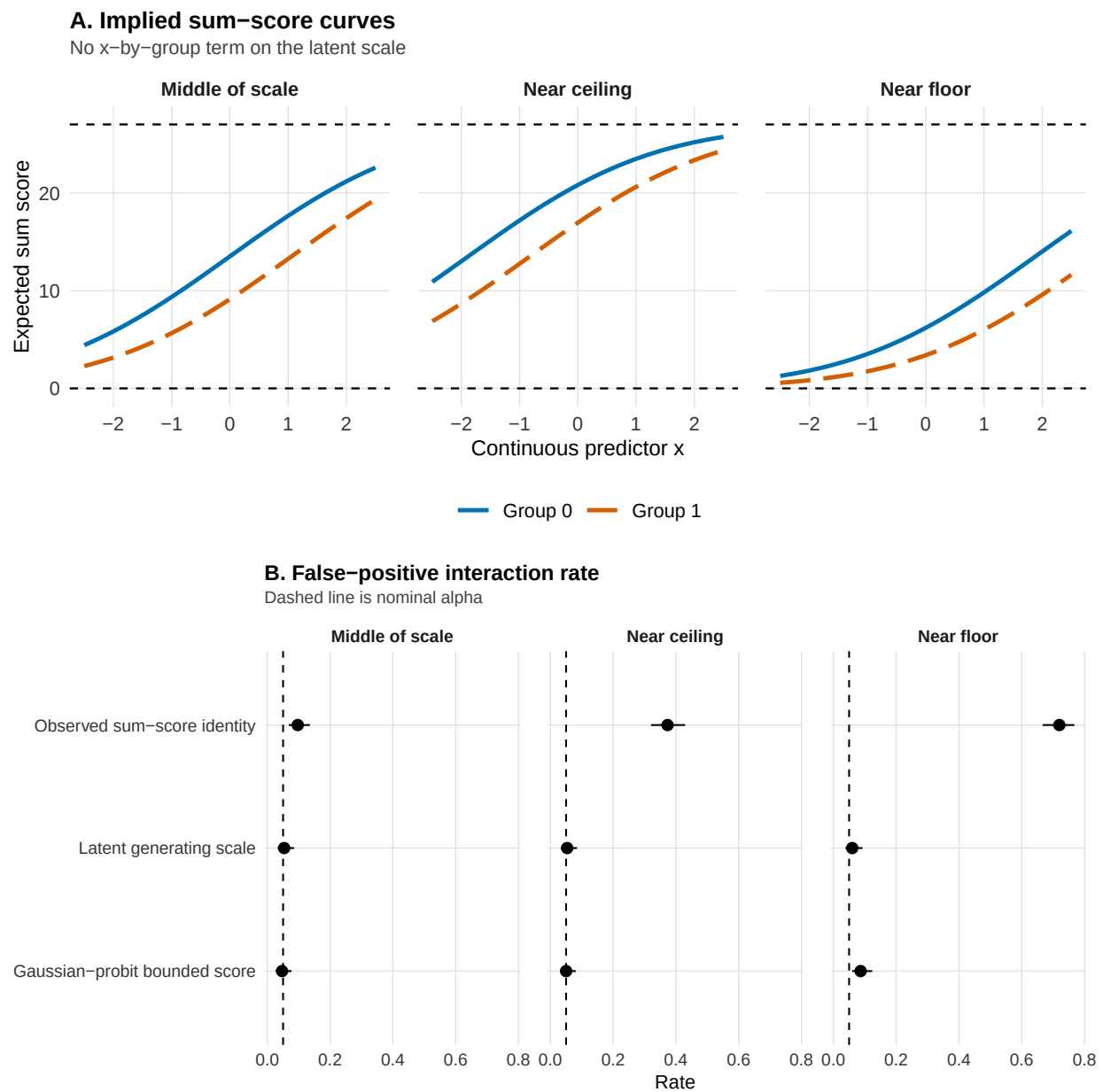
Simulation 3: Design and Results

In each of 300 replications we generated data for $N = 600$ participants on a latent continuous scale, with a continuous predictor, a group effect, and no true x-by-group interaction term. Each latent value was converted into 9 ordinal items scored 0 to 3, which were summed to a total on a 0-to-27 scale. Three scenarios shift the item thresholds so that expected totals fall in the middle of the scale, near the floor, or near the ceiling. Each participant's total score was analyzed on three scales, chosen to isolate the relevant comparison: the observed sum-score identity model, a Gaussian linear model of the manifest total; the Gaussian-probit bounded-score model, a sensitivity model that maps the total to a bounded proportion through a probit link; and the latent generating scale itself, an idealized benchmark available only because the true θ is known. We recorded how often each returned a significant x-by-group interaction term at $\alpha = .05$, a false positive by construction. Figure 5 shows the results; the per-scenario rates are tabulated in the Supplement.

The bounded-score model fits a continuity-corrected score proportion with a Gaussian likelihood and a probit link, then rescales predictions to the sum-score metric. The probit link is more interpretable here than a logit link because its linear predictor is on a standard-normal, z -like scale. In the idealized threshold-normal setting used in the simulation, this makes the coefficient scale closer to standardized latent differences. This interpretation remains conditional on the measurement assumptions and should not be read as a literal Cohen's d for the target construct.

Figure 5

Simulation 3: sum scores as bounded and discrete outcomes. Panel A shows how a latent additive data-generating model, with no x -by-group interaction term, is mapped onto the observed sum-score scale under middle, floor, and ceiling scenarios. Panel B shows false-positive interaction rates across repeated simulations. The latent generating scale is an idealized benchmark available only because θ is known in the simulation; applied analyses have no access to it.



The latent generating scale, the idealized benchmark, stays near the nominal .05 in every scenario (between 5% and 6%), confirming that the interaction is genuinely absent on the scale that generated the data. The observed sum-score identity model matches it only in the middle of the scale, where the total is approximately linear in the latent variable (10%). As expected totals approach a bound, the raw-score model inflates sharply, reaching 37% near the ceiling and 72% near the floor, because uneven compression of the latent scale makes a constant latent group effect appear to change across levels of the predictor. The Gaussian-probit bounded-score model stays close to nominal throughout (between 5% and 9%), showing that respecting the bounds of the score removes most of the inflation.

The pattern is conditional on the region of the score range. Far from the bounds, and for a claim about raw-score units, the Gaussian identity analysis of the total approximates the latent-scale result. Near a bound, the raw-score no-interaction baseline diverges from the latent one, and a interaction term can appear that reflects compression of the metric even when the construct-level effect is constant. The bounded-score probit model is a useful sensitivity check, and it remains a model of the manifest total. When the claim concerns an interaction between latent constructs, the more direct solution is an explicit measurement model, such as SEM, item-level ordinal, or IRT-inspired models; these still require assumptions about thresholds, links, loadings, and latent-response metrics, so they make the measurement part of the interaction claim explicit without removing the scale problem.

What Diagnostics Can and Cannot Detect

Diagnostics belong in link-aware modeling, and by themselves they leave the link-function question undecided. The central problem is diagnostic misalignment: a wrong link can generate a pseudo-interaction that becomes statistically detectable, while the checks used to diagnose the model remain quiet, have limited power, or flag only broad misfit. In that situation, the interaction test can become sensitive to the wrong scale before the diagnostic evidence becomes specific enough to identify its source.

Pregibon-style link tests target link adequacy directly ([Pregibon, 1980](#)). The idea is

simple. The analyst fits the model, saves the fitted linear predictor $\hat{\eta}$, and refits the model with $\hat{\eta}^2$ added as a predictor in this secondary analysis. Under an adequate link, the squared term captures nothing systematic and its test stays near the nominal level; a significant squared term signals curvature that the assumed link leaves unexplained. A significant added term therefore suggests that the assumed link may be inadequate, and it stops short of identifying the correct link: the same signal can reflect a missing nonlinear predictor, an omitted covariate, an omitted interaction, or another problem in the systematic component. A non-significant result is also limited: it means that this diagnostic, in this sample, did not detect this particular departure.

Simulation-based residual diagnostics provide a broader check. The DHARMA package implements randomized quantile residuals (Dunn & Smyth, 1996; Hartig, 2024). The idea is again simple. Many new datasets are simulated from the fitted model, and each observed response is located within the distribution of its own simulated values: an observation at the median of its simulated distribution receives a residual of .50, and an observation in the far tail receives a residual near 0 or 1. When the model is adequate, these positions are uniformly distributed between 0 and 1 for every outcome family, so misfit appears as departures from uniformity, as over- or underdispersion, or as residual trends over fitted values and predictors. These residuals are easier to interpret across GLMs and GLMMs than raw, Pearson, or deviance residuals. In the diagnostic simulation below, we use four DHARMA checks: residual uniformity, dispersion, residual quantile patterns over fitted values, and residual structure over the focal predictor, implemented as a quantile check for continuous predictors and as a design-cell comparison for the categorical 2×2 case. These checks detect broad misfit, dispersion problems, and structured residual patterns. By themselves, they do not distinguish link misspecification from family misspecification, a missing nonlinearity, a missing interaction, or a target-metric mismatch.

Model comparison tools, such as AIC, BIC, WAIC, or LOO, add evidence within a clearly defined candidate set. For link diagnostics, their cleanest use is to compare candidate links while holding fixed the observed outcome, likelihood family or support, and structural formula. In that use, AIC can compare Poisson-log and Poisson-identity fits to the same counts, binomial-logit and

binomial-probit fits to the same successes and trials, or Gamma-log and Gamma-inverse fits to the same positive outcome. It can also compare a standard binomial link with a chance-corrected binomial link when both define likelihoods for the same binomial data. It is less direct when the comparison changes the outcome, the likelihood support, or the measurement metric, which is why we do not treat sum-score alternatives as a simple AIC contest among links. More importantly, information criteria compare predictive fit among the models supplied by the analyst. They do not decide which scale defines the scientific no-interaction claim. A wrong or incomplete link may even be favored when the interaction term absorbs part of the curvature created by the misspecification. Thus, a lower AIC is evidence about relative fit within the candidate set; the adequacy of the link for the scientific claim requires separate justification.

For interaction claims, diagnostics therefore contribute one part of the argument. A useful order is: first identify constraints from the task, scoring rule, and measurement process; then define a small set of plausible outcome families, links, and target metrics; then use residual diagnostics, link checks, calibration, and information criteria as supporting evidence; finally, report whether the interaction conclusion is stable across the plausible specifications. A quiet diagnostic result should not end the assessment when the outcome has known constraints. For example, in a forced-choice task with a known chance level, a standard binomial logit can fit many aspects of the data while still encoding the wrong lower asymptote for the scientific claim. If the claim concerns response-scale moderation, researchers should report response-scale contrasts or marginal effects and examine whether those quantities are stable across plausible links ([McCabe et al., 2022](#); [Mize, 2019](#); [Rohrer & Arel-Bundock, 2026](#)). If the claim concerns additivity on a latent or linear-predictor scale, the chosen link needs a substantive justification.

Diagnostic Simulation

Finally, we examined whether common diagnostics flag the same link problems that produced the pseudo-interactions. The simulation asks a focused question: does diagnostic evidence appear before, after, or at the same time as the false interaction?

We used five scenarios, each tied to a previous example or matched counterpart in the

paper. The first is tied to the simple count example in Figure 1. Data are generated from a Poisson-log model in which expected errors decline with age, with no age-by-group interaction term on the log-count scale. The diagnostic scenario uses the same qualitative structure as the figure, with coefficients recorded in the generated scenario table: intercept 2.20, age effect -0.55 per year after age 6, and group effect 0.45. The fitted model keeps the Poisson family and the interaction formula and uses an identity link. This makes the comparison a same-family link comparison: the family and the formula are held fixed, and only the link changes.

The second scenario is the lower-performance forced-choice scenario from Simulation 1. Data are generated with a .50 chance floor, 20 trials per participant, intercept -0.80, age effect 0.60, group effect -0.90, and no age-by-group interaction term on the chance-corrected logit scale. The fitted model is a standard binomial-logit interaction model, which respects the binomial trial structure while placing the lower asymptote at 0, below the task-implied chance level.

The third scenario mirrors the probit-generated, logit-fitted case in Simulation 2. Data are repeated binary trials with a subject random intercept, 15 trials per condition cell, and latent ICC .30. The fixed effects are the probit reference values used in Simulation 2: intercept 1.50, group effect -1.00, condition effect -1.00, and no group-by-condition interaction term. The fitted model is the corresponding random-intercept logit GLMM with the same interaction formula.

The fourth scenario is a continuous-predictor counterpart to the previous binary repeated-trials case. It keeps the probit data-generating link, the logit fitted link, the group effect, the focal-predictor effect, the subject random intercept, and the latent ICC, and replaces the two-level condition with a continuous predictor sampled from 0 to 1. This scenario asks whether the same logit-versus-probit issue is visible when the focal predictor has enough unique values for continuous-predictor residual checks to be better identified.

The fifth scenario uses the Gamma response-time example from the introduction. Data are generated from a Gamma model with a log link, shape 8, baseline mean 400 when $x = 0$ and group = 0, x effect 0.25 on the log-mean scale, group effect 0.25, and no x -by-group interaction term. The fitted model keeps the Gamma family and the interaction formula and uses the inverse link.

In each replication, diagnostics were applied to the misspecified interaction model, because this is the model an analyst would usually inspect after finding the interaction. This design targets a realistic post-estimation question: does the same fitted model that produced the product-term finding also show enough diagnostic evidence to warn the analyst about the link problem? The choice is demanding because the interaction term can absorb part of the curvature created by the wrong link. A diagnostic workflow applied to an additive model before adding the interaction term would answer a different question. The present simulation asks whether ordinary checks protect the interaction conclusion once the misspecified interaction model has been fitted.

The residual workflow used four DHARMA checks: residual uniformity, dispersion, residual quantile patterns over fitted values, and residual structure over the focal predictor, using a quantile check for continuous predictors and a design-cell comparison for the categorical 2×2 case (Dunn & Smyth, 1996; Hartig, 2024). We also used a Pregibon-style added-term link check as a more link-oriented diagnostic (Pregibon, 1980).

Finally, we used AIC only for relative comparisons among prespecified candidate mean models fitted to the same response data with the same interaction formula. In the count and Gamma scenarios, the response family is held fixed and only the link changes. In the binary scenarios, the binomial sampling model is held fixed. The chance-floor scenario compares two binomial mean mappings: a standard logit model and a chance-corrected logit model with a design-implied lower asymptote. Thus, AIC varies only the candidate mean-link mapping, and the interaction term is present in every candidate model. We report the proportion of replications in which the lower AIC favored the target model. Because no inconclusive ΔAIC category is used, the complement favors the misspecified model when both AIC values are available. These values describe relative fit within a limited candidate set.

Table 3 documents diagnostic misalignment: the same misspecification that generates a pseudo-interaction often does not produce a clear diagnostic signal. In several same-family scenarios, AIC or the Pregibon-style check favored the target model; protection was uneven and depended on which candidate models were compared. The starkest result is the chance-floor

Table 3

Diagnostic simulation across five link-misspecification scenarios. FP interaction reports how often the misspecified interaction model detected a interaction term absent on the generating link scale. DHARMA gives the range across the four prespecified residual checks. Pregibon reports the added-term link check. AIC favors target reports the proportion of replications in which the lower AIC favored the target interaction model over the misspecified interaction model, with the structural formula held fixed. The complementary proportion favors the misspecified model when both AIC values are available. CI = Wilson 95% confidence interval.

Case	Target model	Misspecified model	FP interaction	DHARMA	Pregibon	AIC favors target
Count errors	Poisson log	Poisson identity	0.942 [0.908, 0.964]	0.007-0.906 (4)	0.996	0.996
Chance-floor accuracy	chance-corrected binomial logit	standard binomial logit	0.617 [0.561, 0.670]	0.000-0.070 (4)	0.303	0.000
Binary repeated trials	binomial probit GLMM	binomial logit GLMM	0.080 [0.054, 0.116]	0.000-0.020 (4)	1.000	0.923
Binary continuous predictor	binomial probit GLMM	binomial logit GLMM	0.060 [0.038, 0.093]	0.000-0.007 (4)	1.000	0.960
Gamma mean response time	Gamma log	Gamma inverse	0.287 [0.238, 0.340]	0.000-0.043 (4)	0.303	0.777

scenario: a psychologically common link problem produced a large pseudo-interaction rate while all four DHARMA checks remained near nominal levels and AIC favored the misspecified model.

Practical Starting Points for Link-Aware Interaction Testing

The applied output of the framework is a small set of outcome-specific starting points. These starting points are prompts for the choices that should be visible in a scientific paper: what family respects the observed response, what link defines the scale of additivity, and what sensitivity check is feasible for the design (Table 4). The preferred link should be justified from

theory, task structure, or measurement whenever possible ([Baranger et al., 2023](#); [Castillo et al., 2026](#)). When these considerations do not uniquely determine the link, the interaction should be probed through a planned link-sensitivity analysis. Researchers should fit a set of theoretically defensible alternative specifications, compute the same target response-scale contrast across them, and state whether the interaction conclusion is stable or link-conditional. This shows which part of the conclusion depends on modeling conventions and which part does not. Stable conclusions across plausible specifications are easier to defend. Unstable conclusions should be reported as conditional on the modeling choice, as choices that affect the claim should be visible, justified, and interpretable ([Mustillo et al., 2018](#)).

Discussion

Whether an interaction claim survives plausible link functions alternatives is a question that existing reporting practice rarely answers. Some claims are stable across defensible links and response-scale summaries; others hold only under a modeling choice that is currently left implicit. This scale-dependence of interactions is well known ([Cox, 1984](#); [Loftus, 1978](#); [Wagenmakers et al., 2012](#)), and in GLMs it takes the concrete form developed above, where the link fixes the scale on which effects are additive. The review and simulations reported here examine how often this choice goes unexamined in practice, and how much it can matter when it does.

The review showed that interaction tests are common, link functions are often implicit, and many interaction-testing articles use identity-link analyses on outcomes for which observed-scale additivity is not self-evident. The three simulation studies complement this picture by demonstrating the risk of family and link misspecification; they quantify pseudo-interactions under known conditions, not how often such misspecifications arise in the literature. Forced-choice accuracy offers the clearest case, where the binomial family is correct but a standard link misplaces the lower asymptote. The logit-probit comparison then shows that the problem persists even within a family, between links that look similar in most range. Sum scores extend it further, to the target metric itself, where the manifest score is already a measurement mapping from item responses.

Table 4

Practical starting points for link-aware interaction testing across common psychological outcomes.

Outcome type	Plausible starting point	Minimal sensitivity check
Forced-choice accuracy	Binomial logit/probit; chance-corrected link when the task floor is part of the estimand.	Compare standard logit/probit and chance-corrected links; report convergence and response-scale contrasts.
Binary / proportion outcome	Binomial logit/probit; consider cloglog for asymmetric processes.	Compare logit/probit; inspect calibration and contrasts near bounds or rare-event regions.
Likert / ordinal item	Ordinal logit/probit; metric model only with a clear observed-scale target.	Compare ordinal and metric summaries; report category probabilities or response-scale contrasts.
Sum score / composite	Gaussian identity when many levels stay far from bounds; otherwise consider item-level, ordinal, or bounded-score models.	Inspect bounds, skew, and score concentration; compare with item-level or bounded-score sensitivity models when feasible.
Counts / error counts	Poisson or negative binomial, usually with log link; identity only with an observed-count target.	Check overdispersion and zero inflation; compare count contrasts across plausible count models.
Response time / skewed positive outcome	Lognormal, Gamma log/inverse, Inverse Gaussian, or transformed Gaussian, matched to absolute, proportional, reciprocal, or transformed change.	Compare raw-, log-, and response-scale summaries; inspect tail fit and predicted values.

These results extend previous work showing that interaction effects are often difficult to estimate precisely even when the model form is correct ([Baranger et al., 2023](#); [Castillo et al., 2026](#)). Here, the difficulty is specific to link, which can produce statistically significant interaction terms even when the true interaction is zero on the generating scale. The diagnostic simulation suggests that standard model checks do not reliably reveal when an interaction is driven by link choice. Residual checks, Pregibon-style tests, and information criteria were useful in some settings, but their performance depended on the candidate set. The chance-floor scenario is the clearest failure case: a wrong link produced a large pseudo-interaction while residual checks

showed no sign of misfit and AIC favored the simpler standard link. This is diagnostic misalignment, not uniform diagnostic failure, and it suggests that diagnostics should be treated as evidence about model fit within a candidate set, not as proof that an interaction claim is robust to link choice (Dunn & Smyth, 1996; Hartig, 2024; Pregibon, 1980).

Four limitations bound the scope of the conclusions. First, the review was descriptive and limited to five reference journals across major areas of psychology. Its numerical estimates should therefore be read as reflecting reporting practice in high-standard outlets during the sampled year. A broader mapping across journals, years, and subfields could later support reporting guidance on interaction tests with constrained outcomes. Second, the review coded directly observable reporting features rather than judging, for each article, whether its interaction was scale-dependent enough to be removable. Third, the simulations are mechanism demonstrations aimed at a specific regime, two established main effects that jointly expose observations to link-function curvature. They isolate scale problems under known data-generating processes and do not estimate prevalence or cover all realistic complications. Finally, logit and probit often lead to very similar conclusions, especially away from boundaries, and for prediction, the link is largely a computational convenience, and what matters is forecast accuracy and staying within admissible bounds, such as probabilities between 0 and 1; our claim is that link choice can matter for interaction claims, not that it always does.

Interaction effects are routinely reported as if the choice of modeling scale had no substantive consequences. Link-aware testing corrects this by separating three judgments that are often run together: which outcome family is plausible, which link defines the scale on which additivity holds, and whether the conclusion survives plausible alternatives to that choice. This reframing clarifies what kind of claim researchers can defend. If the result is stable across links, the claim can be about the interaction itself; if it is not, the claim is narrower, but still defensible when stated as an interaction on a particular scale rather than treated as a failure.

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